

1. NAME OF THE MEDICINAL PRODUCT

Fludarabine Sandoz 50 mg/2ml concentrate for solution for injection or infusion.

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

One vial of 2 ml contains 50 mg fludarabine phosphate.

For a full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Concentrate for solution for injection or infusion.

Clear, colourless to almost colourless solution.

The diluents that can be used for dilution are sodium chloride 0.9%.

A clear, colourless solution after dilution.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Treatment of B-cell chronic lymphocytic leukemia (CLL) in patients with sufficient bone marrow reserves.

First line treatment in patients with advanced disease, Rai stages III/IV (Binet stages C), or Rai stages I/II (Binet stages A/B) where the patient has disease-related symptoms or evidence of progressive disease.

4.2 Posology and method of administration

Adults: Fludarabine Sandoz should be administered under the supervision of a qualified physician experienced in the use of antineoplastic therapy.

It is strongly recommended that Fludarabine Sandoz should only be administered intravenously.

No cases have been reported in which paravenously administered Fludarabine Sandoz led to severe local adverse reactions. However, unintentional paravenous administration must be avoided.

The recommended dose is 25 mg fludarabine phosphate/m² body surface given daily for 5 consecutive days every 28 days by the intravenous route. Each vial contained 2 ml solution, each ml of the solution will contain fludarabine phosphate 25 mg (see section 6.5 Special precautions for disposal and other handling).

The required dose (calculated on the basis of the patient's body surface) of the solution is drawn up into a syringe. For intravenous bolus injection, this dose is further diluted into 10 ml of sodium chloride 0.9%. Alternatively, for infusion, the required dose drawn up in a syringe may be diluted into 100 ml of sodium chloride 0.9% and infused over approximately 30 minutes.

The duration of treatment depends on the treatment success and the tolerability of Fludarabine Sandoz.

In CLL patients, Fludarabine Sandoz should be administered up to the achievement of best response (complete or partial remission, usually 6 cycles) and then Fludarabine Sandoz should be discontinued.

Children: The safety and effectiveness of Fludarabine Sandoz in children has not been established. Therefore, Fludarabine Sandoz is not recommended for use in children.

Reduced Kidney Function: Doses should be adjusted for patients with reduced kidney function. If creatinine clearance (CrCl) is between 30 and 70 ml/min, the dose should be reduced by up to 50% and close hematological monitoring should be used to assess toxicity (see section 4.4 Special warnings and precautions for use). Fludarabine Sandoz treatment is contraindicated if creatinine clearance is <30 ml/min (see section 4.3 Contraindications).

Reduced Liver Function: No data are available concerning the use of Fludarabine Sandoz in patients with hepatic impairment. In this group of patients, Fludarabine Sandoz should be used with caution and administered if the perceived benefit outweighs any potential risk.

Elderly Patients: Since there are limited data for the use of Fludarabine Sandoz in elderly persons (>75 years), caution should be exercised with the administration of Fludarabine Sandoz in these patients.

In patients >65 years, creatinine clearance should be measured.

4.3 Contraindications

- Hypersensitivity to the active substance or to any of the excipients
- Renal impairment with creatinine clearance < 30 ml/min
- Decompensated haemolytic anaemia
- Lactation

4.4 Special warnings and precautions for use

Neurotoxicity

When used at high doses in patients with acute leukaemia, intravenous fludarabine phosphate was associated with severe neurologic effects including blindness, coma, and death. Symptoms appeared from 21 to 60 days from last dose. This severe central nervous system toxicity occurred in 36% of patients treated intravenously with doses approximately four times greater (96 mg/m²/day for 5-7 days) than the recommended dose. In patients treated at doses in the range of the dose recommended for chronic lymphocytic leukaemia, severe central nervous system toxicity occurred rarely (coma, seizures, and agitation) or uncommonly (confusion).

Patients should be closely observed for signs of neurologic effects.

The effect of chronic administration of fludarabine phosphate on the central nervous system is unknown. However, patients tolerated the recommended dose, for relatively long treatment times (for up to 26 courses of therapy).

Neurotoxicity has been reported to occur earlier or later.

Impaired state of health

In patients with impaired state of health, fludarabine phosphate should be given with caution and after careful risk/benefit consideration. This applies especially for patients with severe impairment of bone marrow function (thrombocytopenia, anaemia, and/or granulocytopenia), immunodeficiency or with a history of opportunistic infection.

Hepatic impairment

In patients with hepatic impairment fludarabine phosphate should be used with caution because it can cause hepatic toxicity. Fludarabine phosphate should only be administered if the perceived benefit outweighs any potential risk. Such patients should be monitored closely for excessive toxicity and dosage modified or the medicinal product discontinued accordingly (see section 4.2).

Renal impairment

The total body clearance of the principle plasma metabolite 2-F-ara-A shows a correlation with creatinine clearance, indicating the importance of the renal excretion pathway for the elimination of the compound. Patients with reduced renal function demonstrated an increased total body exposure (AUC of 2F-ara-A). There are limited data available in patients with impairment of renal function (creatinine clearance <70 ml/min).

Fludarabine phosphate must be administered cautiously in patients with renal insufficiency. In patients with moderate impairment of renal function (creatinine clearance between 30 and 70 ml/min), the dose should be reduced by up to 50 % and the patient should be monitored closely (see section 4.2).

Fludarabine phosphate treatment is contraindicated if creatinine clearance is < 30 ml/min (see section 4.3).

Myelosuppression

Severe bone marrow suppression, notably anaemia, thrombocytopenia and neutropenia, has been reported in patients treated with fludarabine phosphate. In solid tumour patients, the median time to nadir counts was 13 days (range: 3-25 days) for granulocytes and 16 days (range: 2-32) for platelets. Most patients had haematologic impairment at baseline either as a result of disease or as a result of prior myelosuppressive therapy.

Cumulative myelosuppression may be seen. While chemotherapy-induced myelosuppression is often

reversible, administration of fludarabine phosphate requires careful haematologic monitoring.

Fludarabine phosphate is a potent antineoplastic agent with potentially significant toxic side effects. Patients undergoing therapy should be closely observed for signs of haematologic and non-haematologic toxicity. Periodic assessment of peripheral blood counts is recommended to detect the development of anaemia, neutropenia, and thrombocytopenia.

Several instances of trilineage bone marrow hypoplasia or aplasia resulting in pancytopenia, sometimes resulting in death, have been reported in adult patients. The duration of cytopenia in the reported cases has ranged from approximately 2 months to approximately 1 year. These episodes have occurred both in previously treated or untreated patients.

As with other cytotoxics, caution should be exercised with fludarabine phosphate when further haematopoietic stem cell sampling is considered.

Transfusion-associated graft-versus-host disease

Transfusion-associated graft-versus-host disease (reaction by the transfused immunocompetent lymphocytes to the host) has been observed after transfusion of non-irradiated blood in fludarabine phosphate treated patients. Fatal outcome as a consequence of this disease has been reported with a high frequency. Therefore, to minimize the risk of transfusion-associated graft-versus-host disease, patients who require blood transfusion and who are undergoing, or who have received, treatment with fludarabine phosphate should receive irradiated blood only.

Skin cancer

The worsening or flare up of pre-existing skin cancer lesions as well as new onset of skin cancer has been reported in some patients during or after fludarabine phosphate therapy.

Tumour lysis syndrome

Tumour lysis syndrome has been reported in patients with large tumour burdens. Since fludarabine phosphate can induce a response as early as within the first week of treatment, precautions should be taken in those patients at risk of developing this complication.

Autoimmune disorders

Irrespective of any previous history of autoimmune processes or Coombs test status, life-threatening and sometimes fatal autoimmune phenomena (e.g. autoimmune haemolytic anaemia, autoimmune thrombocytopenia, thrombocytopenic purpura, pemphigus, Evans' syndrome) have been reported to occur during or after treatment with fludarabine phosphate. The majority of patients experiencing haemolytic anaemia developed a recurrence in the haemolytic process after rechallenge with fludarabine phosphate. Patients treated with fludarabine phosphate should be closely monitored for signs of haemolysis.

Patients undergoing treatment with fludarabine phosphate should be closely monitored for signs of autoimmune haemolytic anaemia (decline in haemoglobin linked with haemolysis and positive Coombs test).

Discontinuation of therapy with fludarabine phosphate is recommended in case of haemolysis. Blood transfusions (irradiated, see above) and adrenocorticoid preparations are the most common treatment measures for autoimmune haemolytic anaemia.

The elderly

Since there are limited data for the use of fludarabine phosphate in elderly persons (> 75 years), caution should be exercised with the administration of fludarabine phosphate in these patients.

In patients aged 65 years or older, creatinine clearance should be measured before start of treatment (see renal impairment and section 4.2).

Children and adolescents

No data are available concerning the use of fludarabine phosphate in children and adolescents. Therefore, treatment with fludarabine phosphate in children and adolescents is not recommended.

Pregnancy

Fludarabine phosphate should not be used during pregnancy unless clearly necessary (e.g. life-threatening situation, no alternative safer treatment available without compromising the therapeutic benefit, treatment cannot be avoided). It has the potential to cause fetal harm (see sections 4.6).

Prescribers may only consider the use of Fludarabine phosphate, if the potential benefits justify the potential risks to the foetus.

Women should avoid becoming pregnant while on Fludarabine phosphate therapy.

Women of childbearing potential must be apprised of the potential hazard to the foetus.

Contraception

Women of child-bearing potential or fertile males must take effective contraceptive measures during and at least for 6 months after cessation of therapy (see section 4.6).

Vaccination

During and after treatment with fludarabine phosphate, vaccination with live vaccines should be avoided.

Retreatment options after initial fludarabine treatment

A crossover from initial treatment with fludarabine phosphate to chlorambucil for non-responders to fludarabine phosphate should be avoided because most patients who have been resistant to fludarabine phosphate have shown resistance to chlorambucil.

4.5 Interaction with other medicinal products and other forms of interaction

When using intravenous fludarabine phosphate in combination with pentostatin (deoxycoformycin) for the treatment of refractory chronic lymphocytic leukaemia (CLL), there was an unacceptably high incidence of fatal pulmonary toxicity. Therefore, the use of fludarabine phosphate in combination with pentostatin is not recommended.

Dipyridamole and other inhibitors of adenosine uptake may reduce the therapeutic efficacy of fludarabine phosphate.

A pharmacokinetic medicinal product interaction was observed in CLL and AML (acute myeloid leukaemia) patients during combination therapy with fludarabine phosphate and Ara-C. During use of fludarabine phosphate in combination with cytarabine the intracellular peak concentration and intracellular exposure of Ara-CTP (active metabolite of cytarabine) increased in leukaemic cells. Plasma concentrations of Ara-C and the elimination rate of Ara-CTP were not affected.

4.6 Pregnancy and lactation

Pregnancy

Fludarabine phosphate should not be used during pregnancy unless clearly necessary (e.g. life-threatening situation, no alternative safer treatment available without compromising the therapeutic benefit, treatment cannot be avoided). Fludarabine phosphate has the potential to cause fetal harm. Prescribers may only consider the use of Fludarabine phosphate, if the potential benefits justify the potential risks to the foetus.

Women of child-bearing potential must be apprised of the potential hazard to the foetus.

Both sexually active men and women of childbearing potential must take effective contraceptive measures during and at least for 6 months after cessation of therapy (see section 4.4).

Lactation

It is not known whether this drug or its metabolites are excreted in human milk.

However, there is evidence that fludarabine phosphate and/or metabolites transfer from maternal blood to milk.

Because of the potential for serious adverse reactions to Fludarabine phosphate in breast-fed infants, Fludarabine phosphate is contraindicated in nursing mothers (see section 4.3).

4.7 Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been performed. However, fludarabine phosphate may reduce the ability to drive and use machines, since e.g. fatigue, weakness, visual disturbances, confusion, agitation and seizures have been observed.

4.8 Undesirable effects

The most common adverse events include myelosuppression (neutropenia, thrombocytopenia and anaemia), infection including pneumonia, cough, fever, fatigue, weakness, nausea, vomiting and diarrhoea. Other commonly reported events include chills, oedema, malaise, peripheral neuropathy, visual disturbance, anorexia, mucositis, stomatitis and skin rash.

Serious opportunistic infections have occurred in patients treated with fludarabine. Fatalities as a consequence of serious adverse events have been reported.

System Organ Class MedDRA	Undesirable Effects
Infections and infestations	Very common: Infections/Opportunistic infections (like latent viral reactivation, e.g. progressive multifocal leukoencephalopathy, herpes zoster virus, Epstein-Barr-virus), pneumonia Rare: Lymphoproliferative disorder (EBV-associated)
Neoplasms benign, malignant and unspecified (incl. cysts and polyps)	Common: Myelodysplastic syndrome and acute myeloid leukaemia (mainly associated with prior, concomitant or subsequent treatment with alkylating agents, topoisomerase inhibitors or irradiation)
Blood and lymphatic system disorders	Very common: Neutropenia, anaemia, thrombocytopenia Common: Myelosuppression
Immune system disorders	Uncommon: Autoimmune disorder (including autoimmune haemolytic anaemia, Evan's syndrome, thrombocytopenic purpura, acquired haemophilia, pemphigus)
Metabolism and nutrition disorders	Common: Anorexia Uncommon: Tumour lysis syndrome (including renal failure, metabolic acidosis, hyperkalaemia, urate crystalluria, hyperphosphatemia, hypocalcemia, hyperuricemia, haematuria)
Nervous system disorders	Common: Neuropathy peripheral Uncommon: Confusion Rare: Coma, seizures, agitation Not known: Cerebral haemorrhage
Eye disorders	Common: Visual disturbance Rare: Blindness, optic neuritis, optic neuropathy
Cardiac disorders	Rare: Heart failure, arrhythmia
Respiratory, thoracic and mediastinal disorders	Very common: Cough Uncommon: Pulmonary toxicity (including pulmonary fibrosis, pneumonitis, dyspnoea) Not known: Pulmonary haemorrhage
Gastrointestinal disorders	Very common: Vomiting, diarrhoea, nausea Common: Stomatitis Uncommon: Gastrointestinal haemorrhage, pancreatic enzymes abnormal
Hepatobiliary disorders	Uncommon: Hepatic enzyme abnormal
Skin and subcutaneous tissue disorders	Common: Rash Rare: Skin cancer, necrolysis epidermal toxic (Lyell type), Stevens-Johnson Syndrome
Renal and urinary disorder	Not known: Haemorrhagic cystitis

General disorders and administration site conditions	Very common: Fever, fatigue, weakness Common: Oedema, mucositis, chills, malaise
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4.9 Overdose

High doses of fludarabine phosphate have been associated with irreversible central nervous system toxicity characterised by delayed blindness, coma, and death. High doses are also associated with severe thrombocytopenia and neutropenia due to bone marrow suppression.

There is no specific antidote for fludarabine phosphate overdosage. Treatment consists of drug discontinuation and supportive therapy.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Antineoplastic agents, ATC code: L01B B05

Fludarabine phosphate contains fludarabine phosphate (2F-Ara-AMP), a water-soluble fluorinated nucleotide analogue of the antiviral agent vidarabine (Ara-A, 9-β-D-arabinofuranosyladenine) that is relatively resistant to deamination by adenosine deaminase.

Fludarabine phosphate is rapidly dephosphorylated to 2F-ara-A, which is taken up by cells and then intracellularly phosphorylated by deoxycytidine kinase to active triphosphate, 2F-ara-ATP. This metabolite inhibits ribonucleotide reductase, DNA polymerase α/β and γ, DNA primase and DNA ligase thereby preventing DNA synthesis. Furthermore, partial inhibition of RNA polymerase II and consequent reduction in protein synthesis occur.

While some aspects of the mechanism of action of 2F-ara-ATP are as yet unclear it is believed that effects on DNA, RNA and protein synthesis all contribute to the inhibition of cell growth, with inhibition of DNA synthesis being the dominant factor. In addition, exposure of CLL lymphocytes to 2F-ara-A has shown triggers extensive DNA fragmentation and cell death characteristic of apoptosis.

5.2 Pharmacokinetic properties

Plasma and urine pharmacokinetics of fludarabine (2F-ara-A)

The pharmacokinetic properties of fludarabine phosphate were investigated after intravenous bolus injection and after short and long term intravenous infusion of fludarabine phosphate (2F-ara-AMP) in patients with malignant diseases.

Distribution and Biotransformation

2F-ara-AMP is a water-soluble prodrug, which is rapidly and quantitatively dephosphorylated in the human organism to the nucleoside fludarabine (2F-ara-A).

After cancer patients had received a single infusion of 25 mg 2F-ara-AMP per m² to over 30 minutes, 2F-ara-A reached mean maximum concentrations in the plasma of 3.5-3.7 μM at the end of the infusion.

Corresponding 2F-ara-A levels after the fifth dose showed a moderate accumulation with mean maximum levels of 4.4-4.8 μM at the end of the infusion. During a 5-day treatment period, 2F-ara-A plasma trough levels increased by a factor of about 2.

An accumulation of 2F-ara-A over several treatment cycles can be ruled out. Post maximum levels of 2F-ara-A were reduced in three disposition phases with an initial half-life of approx. 5 minutes, an intermediate half-life of 1-2 hours and a terminal half-life of approx. 20 hours.

Occurrence of neutropenia and haematocrit changes indicated that cytotoxicity of fludarabine phosphate causes dose dependent haematopoiesis inhibition.

Elimination

The main pathway of 2F-ara-A elimination leads via the kidneys. 40 to 60% of the administered IV dose was excreted in the urine. In animals with ³H-2F-ara-AMP showed that the radio-tagged substances were completely excreted in the urine. 2F-ara-hypoxanthine, which is the major metabolite in dogs, was observed in

humans only in small amounts. Since patients with impaired renal function have a reduced total body clearance of 2F-ara-A, dose reduction is required in these cases. Human plasma proteins showed no pronounced tendency of 2F-ara-A protein binding.

Cellular pharmacokinetics of fludarabine triphosphate

2F-ara-A is actively absorbed into leukemic cells, whereupon it is rephosphorylated to mono- and diphosphate and subsequently to triphosphate. Fludarabine triphosphate, 2F-ara-ATP, is the major intracellular metabolite and the only one known to have cytotoxic activity. Maximum 2F-ara-ATP levels in leukemic lymphocytes of CLL patients were observed approx. 4 hours after application and exhibited a considerable variation around a median peak concentration of approx. 20 µM. 2F-ara-ATP levels in leukemic cells were considerably higher than maximum 2F-ara-A plasma levels indicating an accumulation in the target cells. Incubation of leukemic lymphocytes showed a linear relationship between extracellular 2F-ara-A exposure (product of 2F-ara-A concentration and incubation time) and intracellular 2F-ara-ATP enrichment. 2F-ara-ATP elimination from target cells showed median half-life values of 15 and 23 hours.

No clear correlation was found between 2F-ara-A pharmacokinetics and treatment efficacy in cancer patients.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Disodium phosphate dihydrate
Sodium hydroxide
Water for injection

6.2 Incompatibilities

This medicinal product must not be mixed with other medicinal products except those mentioned in section 6.5.

6.3 Shelf life

As packaged for sale:

3 years.

Shelf life after dilution:

Infusion solutions detailed in section 6.5 are physically and chemically stable for 7 days when stored in a refrigerator (2-8°C) with protection from light and at room temperature (20°C-25°C) with and without protection from light.

From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would normally not be longer than 24 hours at 2°C to 8°C, unless dilution has taken place in controlled and validated aseptic conditions.

6.4 Special precautions for storage

As packaged for sale:

Store in a refrigerator (2°C – 8°C).

For storage condition of the diluted medicinal product, see section 6.3.

6.5 Special precautions for disposal and other handling

Dilution

The required dose (calculated on the basis of the patient's body surface) is drawn up into a syringe.

For intravenous bolus injection, this dose is further diluted in 10 ml of sodium chloride 9 mg/ml (0.9%) solution for injection. Alternatively, for infusion, the required dose may be diluted in 100 ml of sodium chloride 9 mg/ml (0.9%) solution for injection and infused over approximately 30 minutes.

Inspection prior to use

Only clear and colourless solutions without particles should be used. The product should not be used in case of a defective container.

Handling and disposal

Pregnant women should be excluded from handling fludarabine phosphate. The regulations concerning proper handling and disposal must be followed considering guidelines for proper handling and disposal of cytotoxic medicinal products. Any spilled or unused material may be eliminated by incineration.

Caution should be exercised during handling and preparation of fludarabine phosphate solution. The use of protective gloves and safety glasses is recommended to avoid exposure in case of breakage of the vial or other accidental spillage. If the solution comes in contact with skin or mucous membranes, the affected area should be cleaned thoroughly with soap and water.

6.6 Packaging

Available pack sizes are 1 vial x 2 ml and 5 vials x 2 ml.

Not all pack sizes may be marketed.

7. PRODUCT REGISTRATION HOLDER

Sandoz Products Malaysia Sdn. Bhd.
Unit 1202, Level 12, Uptown 1,
No. 1, Jalan SS21/ 58, Damansara Uptown,
47400 Petaling Jaya Selangor, Malaysia

8. DATE OF REVISION OF THE TEXT

Mar 2024